PHASE - 3

1. Schema Overview

Purpose of the Database

The Avento database serves as the central repository for storing and managing all information related to university event management activities. It supports user authentication, role-based access control, event creation and management, participant registrations, notifications, and other administrative operations. The database ensures secure storage, efficient retrieval, and consistent management of data across the platform.

Database Type

Avento utilizes Supabase as its backend database service, which is built on PostgreSQL. The database follows a relational model, enabling structured data storage, strong consistency, referential integrity, and efficient querying through well-defined relationships between tables.

Design Objectives

The database schema has been designed with the following objectives:

- Ensure secure and reliable storage of user and event information.
- Support role-based access control for students and administrators.
- Maintain data integrity through primary key and foreign key relationships.
- Prevent duplicate registrations and unauthorized access.
- Enable efficient event discovery, registration, and management.
- Support scalable growth in users, events, and participation records.
- Facilitate real-time updates and notifications through Supabase services.
- Provide a maintainable and extensible structure for future feature enhancements.

The schema follows a normalized relational design to minimize data redundancy while ensuring efficient access to frequently used information. Appropriate constraints and relationships are implemented to maintain consistency and enforce business rules throughout the system.

2. Schema Design – High Level Architecture

Overview

The Avento database follows a relational architecture implemented using Supabase (PostgreSQL). The schema is organized into multiple interconnected tables, each responsible for managing a specific aspect of the system. Relationships between tables are established

through primary keys and foreign keys, ensuring data integrity and consistency across the platform.

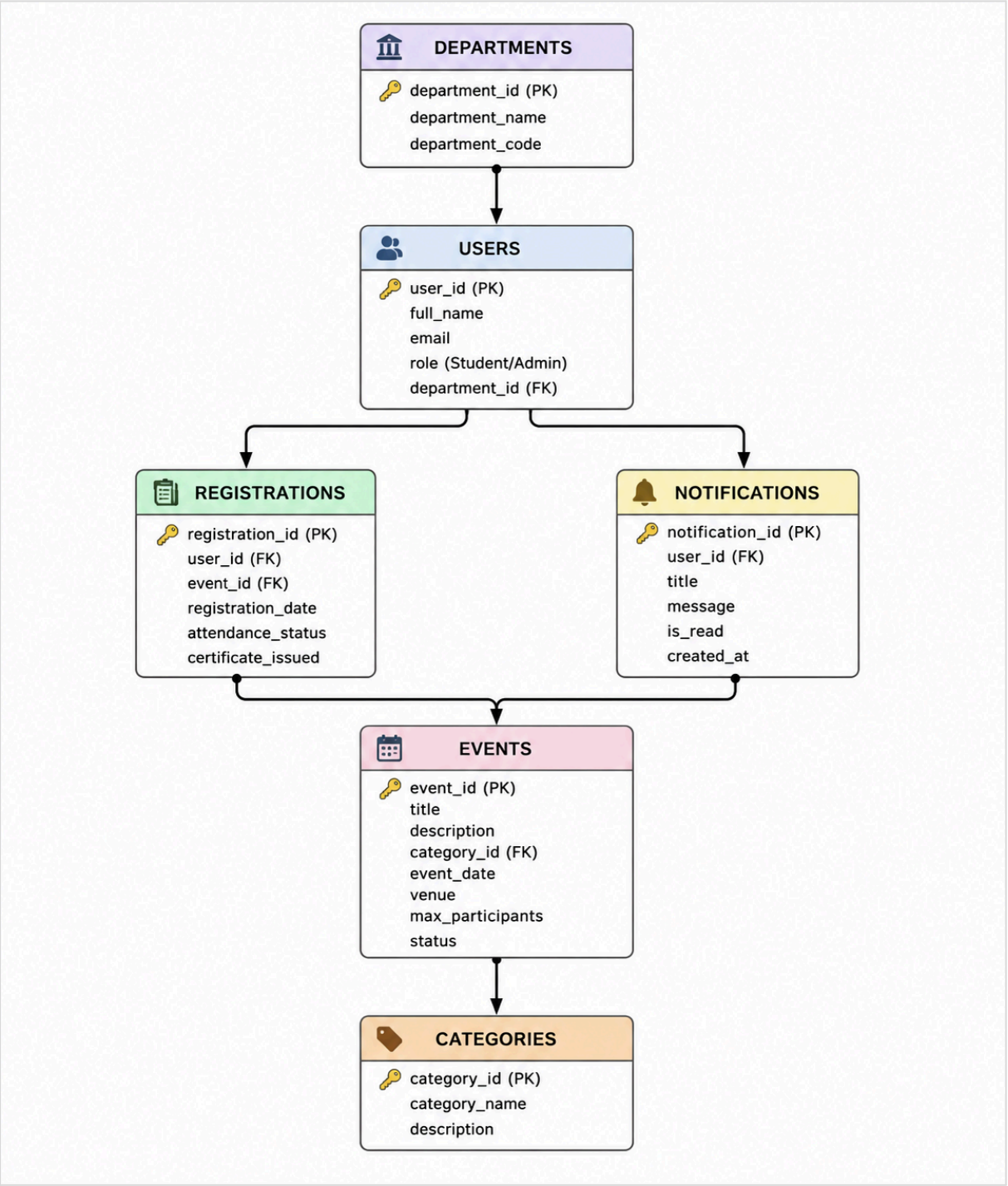
The database architecture is designed to support secure user authentication, event management, participant registrations, notification delivery, and administrative operations while maintaining scalability and efficient data access.

Core Tables

Table Name	Purpose
Users	Stores user account information including student/admin details, authentication data, role information, and verification status.
Departments	Maintains academic department information and associates users with their respective departments.
Categories	Stores event categories such as Technical, Cultural, Sports, Workshops, Seminars, and other classifications.
Events	Contains event-related information including title, description, venue, schedule, participant limits, and event status.
Registrations	Records student registrations for events and serves as the relationship table between users and events.
Notifications	Stores notifications, announcements, reminders, and event-related updates delivered to users.
OTP_Verifications	Maintains temporary OTP records used for account verification and authentication processes.
User_Sessions	Stores active session information and authentication tokens for secure session management.

The interaction between these tables enables the complete functioning of the Avento platform, supporting user management, event organization, participant registration, communication, and security-related operations.

High-Level Schema Architecture



Interaction Flow

1. Users belong to a specific department and are assigned system roles such as Student or Administrator.
2. Administrators create and manage events that are classified under predefined event categories.
3. Students browse available events and register through the registration module.

4. Registration records establish the relationship between users and events while preventing duplicate registrations.
5. Notifications are generated to inform users about event announcements, updates, registration confirmations, and other system activities.
6. OTP verification records support secure account creation and authentication processes during user onboarding.

The architecture ensures clear separation of responsibilities among tables while maintaining strong relationships that support efficient data management and future system expansion.

3. Table-by-Table Schema Definitions

3.1 Users Table

The **Users** table stores information about all registered users of the Avento platform, including students and administrators. It serves as the primary table for user authentication, authorization, and profile management.

Attributes

Attribute	Description
user_id	Primary Key
full_name	User's full name (Required)
email	Unique email address (Required)
password	Encrypted password (Required)
role	User role (Student/Admin)
department_id	Foreign Key referencing Departments
year_of_study	Academic year of the student
is_verified	Account verification status (Default: False)
created_at	Account creation timestamp

Relationships

- One Department can have many Users.
- One User can have many Registrations.
- One User can receive many Notifications.

3.2 Departments Table

The **Departments** table stores information about academic departments within the institution and is used to categorize users according to their department affiliation.

Attributes

Attribute	Description
department_id	Primary Key
department_name	Name of the department (Required)
department_code	Unique department code

Relationships

- One Department can be associated with multiple Users.

3.3 Categories Table

The **Categories** table stores event classifications such as Technical, Cultural, Sports, Workshops, and Seminars. These categories help organize and filter events within the platform.

Attributes

Attribute	Description
category_id	Primary Key
category_name	Category name (Required)

Relationships

- One Category can contain multiple Events.

3.4 Events Table

The **Events** table stores detailed information about events created and managed through the platform. It acts as the central table for event management operations.

Attributes

Attribute	Description
event_id	Primary Key
title	Event title (Required)
description	Event description (Required)
category_id	Foreign Key referencing Categories
organizer_id	Foreign Key referencing Users (Admin)
venue	Event venue (Required)
event_date	Scheduled event date (Required)
start_time	Event start time (Required)
end_time	Event end time (Required)
registration_deadline	Last date for registration (Required)
max_participants	Maximum participant limit
status	Event status (Upcoming, Ongoing, Completed, Cancelled)

Relationships

- One Category can have many Events.
- One Administrator can create many Events.
- One Event can have many Registrations.

3.5 Registrations Table

The **Registrations** table records student participation in events and establishes the relationship between users and events.

Attributes

Attribute	Description
registration_id	Primary Key
user_id	Foreign Key referencing Users
event_id	Foreign Key referencing Events

registration_date	Registration timestamp
attendance_status	Attendance status (Default: False)
certificate_issued	Certificate issuance status (Default: False)

Relationships

- Many Registrations belong to one User.
- Many Registrations belong to one Event.

Constraints

- A user can register only once for a specific event.
- Registration is allowed only before the registration deadline.
- Event participant limits must not be exceeded.

3.6 OTP Verification Table

The **OTP Verification** table temporarily stores One-Time Password (OTP) information used during account registration and verification.

Attributes

Attribute	Description
otp_id	Primary Key
email	User email address (Required)
otp_code	Generated OTP code (Required)
expires_at	OTP expiration timestamp
verified	Verification status (Default: False)

Purpose

This table supports secure account verification by validating institutional email ownership before granting access to the platform.

3.7 Notifications Table

The **Notifications** table stores announcements, reminders, event updates, and system-generated messages delivered to users.

Attributes

Attribute	Description
notification_id	Primary Key
user_id	Foreign Key referencing Users
title	Notification title (Required)
message	Notification content (Required)
is_read	Read status (Default: False)
created_at	Notification creation timestamp

Relationships

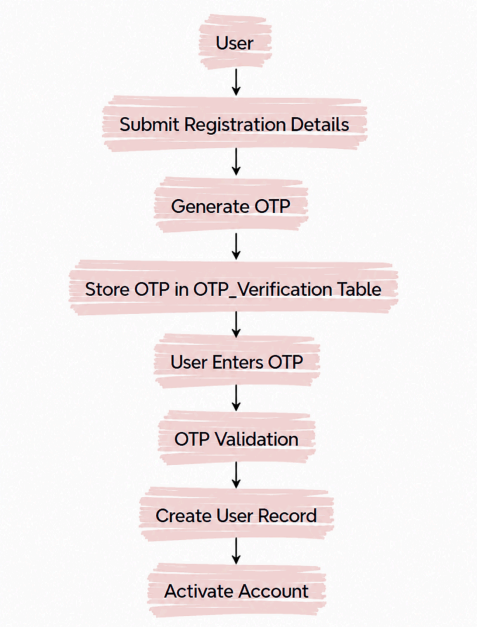
- One User can receive multiple Notifications.

4. Key Database Flows

The Avento database supports several core operational workflows that facilitate user management, event administration, participant registration, and system communication. These workflows illustrate how data moves between tables and how different components of the system interact during key operations.

4.1 User Registration and Verification Flow

This workflow handles the onboarding of new users into the platform. Users must verify their institutional email address through OTP authentication before their account becomes active.



Tables Involved

- Users
- OTP_Verification

4.2 Event Creation Flow

This workflow enables administrators to create and publish events that can be viewed and registered for by students.



Tables Involved

- Events
- Categories
- Notifications
- Users

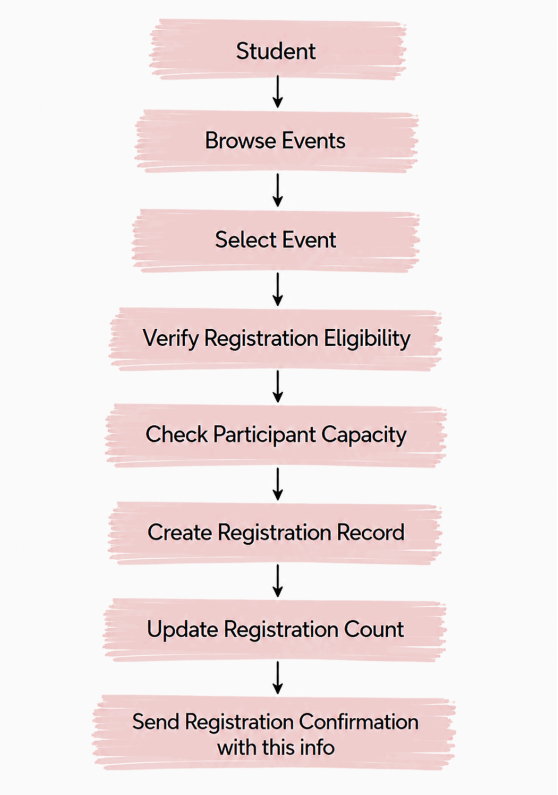
4.3 Event Registration Flow

This workflow manages student participation in events and ensures that registration constraints are enforced before a registration record is created.

Tables Involved

- Users

- Events
- Registrations
- Notifications

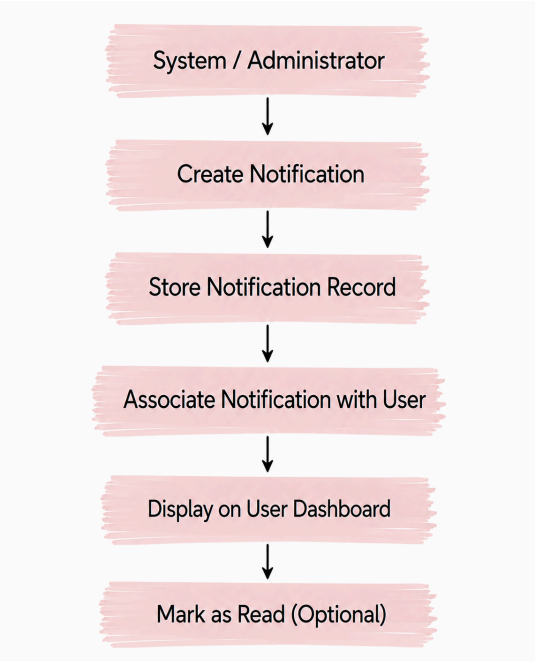


4.4 Notification Management Flow

This workflow manages the generation and delivery of notifications related to events, announcements, reminders, and platform activities.

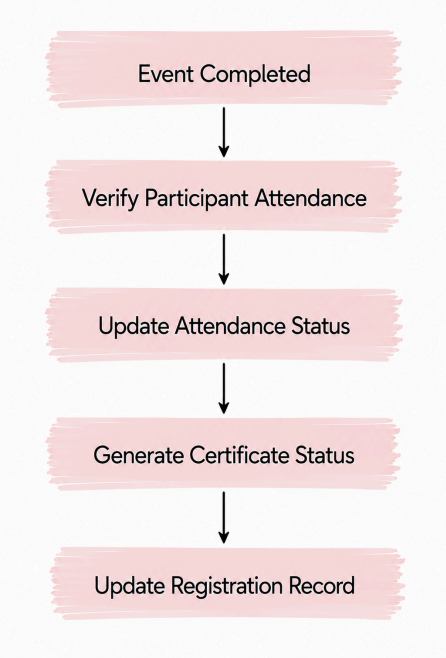
Tables Involved

- Notifications
- Users



4.5 Event Participation Tracking Flow

This workflow records attendance and certificate status after the completion of an event.



Tables Involved

- Events
- Registrations
- Users

The above workflows represent the primary database interactions within the Avento platform and demonstrate how information is processed, stored, and maintained throughout the system lifecycle.

Note: The database schema presented in this document represents the logical structure of the Avento platform. The schema is implemented using Supabase (PostgreSQL) and is designed to ensure data integrity, scalability, security, and efficient management of users, events, registrations, and notifications. The design may be extended in future versions to support additional features and system enhancements.

5. Sample SQL Table Definitions

To demonstrate how the database schema can be implemented in Supabase (PostgreSQL), sample SQL table definitions are provided below. These examples illustrate the use of primary keys, foreign keys, constraints, and default values within the Avento database.

5.1 Users Table

```
CREATE TABLE Users (
   user_id UUID PRIMARY KEY,
   full_name VARCHAR(100) NOT NULL,
   email VARCHAR(255) UNIQUE NOT NULL,
   password VARCHAR(255) NOT NULL,
   role VARCHAR(20) NOT NULL,
   department_id UUID,
   year_of_study INTEGER,
   is_verified BOOLEAN DEFAULT FALSE,
   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5.2 Events Table

```
CREATE TABLE Events (
   event_id UUID PRIMARY KEY,
   title VARCHAR(255) NOT NULL,
   description TEXT NOT NULL,
   venue VARCHAR(255) NOT NULL,
   event_date DATE NOT NULL,
```

```
    max_participants INTEGER,
    status VARCHAR(20),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Purpose

These sample definitions demonstrate the structural implementation of the Avento database. In the actual system, additional constraints, foreign key relationships, validation rules, and security policies are implemented through Supabase to ensure data integrity and secure access control.